

Objective: support my provisional claim that I'm the *best applicant* for the Senior Software Engineer, Safety

Professional
experience

Sabbatical

February 2021 - present

- Extended leave for various family caregiving
- Open source passion project (Python+Scala) **2023-present**
 - Agentic **voice assistant layer** on top of a plaintext markdown wiki
 - OpenAI Whisper for voice transcription
 - Rasa open source for ML
 - "Voting" inspired by Monty (Thousand Brains Project)
 - Distributed computing using: Akka 2.6, Pykka and mqtt
 - Like Livongo's async backend architecture but "atomic" notes instead of microservices
 - [[Notification Center]] Obsidian pane for **human-in-the-loop**
 - **Audit trails** for nondeterministic agents in workflows
 - IoT integrations: Aranet4, AirGradient, PurpleAir, smart lights, etc
 - Network visualization using D3 in Typescript

Backend and Data "Hybrid" Engineer	March 2018 - February 2021
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Livongo (now Teledoc), Mountain View, CA (Hybrid)

Accomplishments:

- Championed documentation in the shared Confluence wiki
 - Essential for cross-team collaboration, reduced context-switching
- Reduced various ETL stage runtimes by up to 50%
- Mentor to one summer intern

Primary Technologies

- Data: Bash, Python, Redshift for SQL, [Jenkins for pipeline orchestration](#)
 - ETL from Mongo to Redshift ([company blog post by me](#))
- Backend team: Play Scala and MySQL, Kafka for async microservices

Software Engineer - Backend	August 2016 - March 2018
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Tile (now Life360), San Mateo, CA

- Implemented Facebook login (used for ~25% of signups), notifications center, in-app promotions, etc.
- Introduced dependency injection (HK2) for faster testing
- Mentor to one summer intern

Technologies

- Jersey (RESTful Web Services in Java - implements JAX-RS)
- [MySQL NDB Cluster](#), *limited* InnoDB
- Jenkins for CI/CD

Software Engineer	July 2013 - July 2016
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LinkedIn, Mountain View, CA

[Student App](#) (*retired* - merged with primary job-search app)

- **Play Scala** and Android

[Original Alumni tool](#)

- Rewrote existing Ruby webapp in **Java**, increased speed and scalability

[University Pages](#)

- Wrote a multithreaded Python script as a workaround to enable a legacy system to ingest tens of millions of schools' data **in time for our launch**

Education

University of California, Davis (Davis, CA)

Bachelor of Science, Computer Science

September 2009 - June 2013

- Web Master to Computer Science Club
 - Lead whiteboard coding interview and code review practices sessions
- Linux Users Group of Davis